Test Series No.

1.7

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

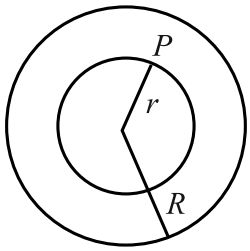
DATE: 26/02/2026

ANSWER KEY

1.	(3)	46.	(1)	91.	(1)	136.	(4)
2.	(4)	47.	(4)	92.	(3)	137.	(3)
3.	(3)	48.	(4)	93.	(1)	138.	(4)
4.	(2)	49.	(3)	94.	(4)	139.	(1)
5.	(3)	50.	(1)	95.	(2)	140.	(2)
6.	(2)	51.	(4)	96.	(3)	141.	(3)
7.	(2)	52.	(4)	97.	(1)	142.	(4)
8.	(2)	53.	(1)	98.	(3)	143.	(2)
9.	(2)	54.	(2)	99.	(2)	144.	(3)
10.	(1)	55.	(3)	100.	(3)	145.	(1)
11.	(4)	56.	(3)	101.	(2)	146.	(2)
12.	(3)	57.	(3)	102.	(2)	147.	(4)
13.	(1)	58.	(1)	103.	(1)	148.	(4)
14.	(1)	59.	(1)	104.	(3)	149.	(1)
15.	(1)	60.	(4)	105.	(3)	150.	(2)
16.	(4)	61.	(2)	106.	(1)	151.	(3)
17.	(1)	62.	(2)	107.	(2)	152.	(1)
18.	(3)	63.	(2)	108.	(1)	153.	(2)
19.	(2)	64.	(2)	109.	(3)	154.	(2)
20.	(1)	65.	(1)	110.	(4)	155.	(3)
21.	(3)	66.	(4)	111.	(1)	156.	(4)
22.	(4)	67.	(2)	112.	(1)	157.	(1)
23.	(2)	68.	(4)	113.	(4)	158.	(2)
24.	(3)	69.	(3)	114.	(3)	159.	(2)
25.	(1)	70.	(4)	115.	(3)	160.	(1)
26.	(1)	71.	(1)	116.	(1)	161.	(3)
27.	(4)	72.	(2)	117.	(2)	162.	(3)
28.	(4)	73.	(2)	118.	(3)	163.	(4)
29.	(3)	74.	(3)	119.	(2)	164.	(1)
30.	(2)	75.	(2)	120.	(1)	165.	(4)
31.	(4)	76.	(4)	121.	(4)	166.	(2)
32.	(1)	77.	(4)	122.	(1)	167.	(3)
33.	(1)	78.	(3)	123.	(1)	168.	(4)
34.	(1)	79.	(4)	124.	(1)	169.	(1)
35.	(2)	80.	(2)	125.	(2)	170.	(3)
36.	(3)	81.	(4)	126.	(2)	171.	(2)
37.	(4)	82.	(2)	127.	(2)	172.	(1)
38.	(1)	83.	(2)	128.	(1)	173.	(1)
39.	(2)	84.	(1)	129.	(4)	174.	(2)
40.	(1)	85.	(1)	130.	(4)	175.	(3)
41.	(3)	86.	(2)	131.	(3)	176.	(1)
42.	(4)	87.	(2)	132.	(2)	177.	(3)
43.	(1)	88.	(2)	133.	(1)	178.	(3)
44.	(3)	89.	(4)	134.	(2)	179.	(3)
45.	(2)	90.	(1)	135.	(4)	180.	(2)

[Contd...

1. (3)



The Electric field at the point P distant r_1 from the centre , according to the gauss theorem is

$$E \cdot 4\pi r_1^2 = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

$$E \cdot 4\pi r_1^2 = \frac{1}{\epsilon_0}\int \rho dV$$

$$\Rightarrow E \cdot 4\pi r_1^2 = \frac{1}{\epsilon_0}\int_0^{r_1} \frac{Qr}{\pi R^4} \cdot 4\pi r^2 dr$$

$$\Rightarrow E = \frac{Qr_1^2}{4\pi\epsilon_0 R^4}$$

(NCERT Ed. 2025-26, 12th, Page No. 35)

2. (4)

As a neutron has more rest mass than a proton, so decay spontaneously releasing energy, while free proton requiring energy input for such conversion.

(NCERT Ed. 2025-26, 12th, Page No. 308)

3. (3)

$v = f\lambda = (4\times10^{14})(5\times10^{-7}) = 2\times10^8$ m/s.

$\mu = c/v = (3\times10^8)/(2\times10^8) = 1.5$.

(NCERT Ed. 2025-26, 12th, Page No. 259)

4. (2)

As in case of linear rheostat R $\propto$ length (l)

$ac = \frac{1}{4}ab$ and resistance of rheostat = 12000 Ω

$\therefore R_{ac} = \frac{1}{4}\times12000 = 3000\Omega$

$R_{bc} = 12000 - 3000 = 9000\Omega$

Resistance of voltmeter, $R_V = 6000\Omega$

The equivalent resistance of voltmeter and part ac connected in parallel is

$$R' = \frac{R_{ac}\times R_V}{R_{ac}+R_V} = \frac{3000\times6000}{3000+6000} = 2000\Omega$$

Resistances \$R'\$ and R_{bc} are in series, therefore net resistance in circuit,

$R = 9000 + 2000 = 11000\Omega$

Current in circuit, $i = \frac{V}{R} = \frac{220}{11000} = 0.02$ A.

$\therefore$ Reading of voltmeter = p.d. across ac

$= iR = 0.02\times2000 = 40$ V

(NCERT Ed. 2025-26, 12th, Page No. 104)

5. (3)

$$I_d = C\left(\frac{dV}{dt}\right) = (20\times10^{-12})\times(2\times10^{11}) = 4 \text{ A.}$$

(NCERT Ed. 2025-26, 12th, Page No. 203)

6. (2)

Einstein's equation: $V = \left(\frac{h}{e}\right)v - \frac{\phi}{e}$

Comparing with $y = mx + c$: Slope $(m) = \frac{h}{e}$

Thus, $h = \text{Slope}\times\text{electron charge}(e)$.

(NCERT Ed. 2025-26, 12th, Page No. 282)

7. (2)

Total Resistance $R_{eq} = 2+3 = 5\Omega$. Power

$$P = \frac{V^2}{R_{eq}} = \frac{(10)^2}{5} = \frac{100}{5} = 20 \text{ W.}$$

(NCERT Ed. 2025-26, 12th, Page No. 108)

8. (2)

When either or both of A & B is closed we get current bypass from switch

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Hence it is "NOR" gate

(NCERT Ed. 2025-26, 12th, Page No. 343)

9. (2)

Resonance occurs at

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50\times10^{-3}\times20\times10^{-6}}} = \frac{1}{\sqrt{10^{-6}}}$$

= 1000 rad/s.

(NCERT Ed. 2025-26, 12th, Page No. 189)

[Contd...

10. (1)

Angular width of the central bright is $\frac{2\lambda}{d}$

$=\frac{2\times6328\times10^{-10}}{0.2\times10^{-3}}\times\frac{180^{\circ}}{\pi}=0.36$ degree

(NCERT Ed. 2025-26, 12th, Page No. 268)

11. (4)

Source current $I_s=(50-10)/2k\Omega=20$ mA.

Load current $I_L=10/2k\Omega=5$ mA.

Zener current $I_z=I_s-I_L=20-5=15$ mA.

(NCERT Ed. 2025-26, 12th, Page No. 340)

12. (3)

For a plano-convex lens, the focal length is given by: $f=\frac{R}{\mu-1}$

Given: $R=10$ cm and $f=30$ cm

$30=\frac{10}{\mu-1}\Rightarrow\mu-1=\frac{10}{30}=0.33$

$\Rightarrow\mu=1+0.33=1.33$

(NCERT Ed. 2025-26, 12th, Page No. 234)

13. (1)

Using the Right-Hand Rule for a moving positive charge:

Point fingers of your right hand in the direction of velocity ($+y$ axis); Curl fingers towards the observation point ($+z$ axis) then the thumb points in the direction of the magnetic field i.e. along the $+x$ axis.

Mathematically: $\vec{B}\propto(\vec{v}\times\vec{r})=\hat{j}\times\hat{k}=+\hat{i}$ ($+x$ axis).

(NCERT Ed. 2025-26, 12th, Page No. 119)

14. (1)

Formula for angular width of central maximum:

$\theta=\frac{2\lambda}{d}$ Where:

- $\lambda=$ Wavelength (depends on frequency)
- $d=$ Slit width

Since the formula only contains λ and d, the angular width is independent of the distance between the slit and the source.

(NCERT Ed. 2025-26, 12th, Page No. 268)

15. (1)

$\lambda=\frac{1240}{E_g}=\frac{1240}{2.48}=500$ nm. 500 nm corresponds to the Green region of the spectrum.

(NCERT Ed. 2025-26, 12th, Page No. 344)

16. (4)

During charging, the potential difference is greater than emf.

During discharging, the potential difference is less than emf.

(NCERT Ed. 2025-26, 12th, Page No. 110)

17. (1)

$\mu=\frac{B}{H}=\frac{0.6}{2000}=3\times10^{-4}$ T · m/A.

(NCERT Ed. 2025-26, 12th, Page No. 146)

18. (3)

Magnetic field on the equatorial line (perpendicular to axis) of a small bar magnet at distance r is: $B=\frac{\mu_0 M}{4\pi r^3}\Rightarrow B\propto\frac{1}{r^3}$

Given distances $r_A=x$ and $r_B=3x$:

$\frac{B_A}{B_B}=\left(\frac{r_B}{r_A}\right)^3=\left(\frac{3x}{x}\right)^3=27$

Ratio = 27: 1

(NCERT Ed. 2025-26, 12th, Page No. 140)

19. (2)

Conservation of Mass Number:

$A+1=4+11\Rightarrow A=14$. Conservation of Charge:

$Z+0=2+5\Rightarrow Z=7$. Nuclide is ${}^{14}_{7}\text{N}$.

(NCERT Ed. 2025-26, 12th, Page No. 315)

20. (1)

Using the Right-Hand Rule ($\vec{F}=q(\vec{v}\times\vec{B})$) where $\vec{B}$ is into the page:

A: Moves down, deflects right $\rightarrow$ Matches RHR $\rightarrow$ Positive (I is Correct)

B: Moves up, deflects right $\rightarrow$ Opposite to RHR (left) $\rightarrow$ Negative (II is Correct)

[Contd...

C: Clockwise circle $\rightarrow$ Centripetal force is opposite to RHR (outward) $\rightarrow$ Negative (III is Correct)

D: Straight line $\rightarrow$ No deflection $\rightarrow$ Neutral (IV is Incorrect)

(NCERT Ed. 2025-26, 12th, Page No. 112)

21. (3)

In β^- decay, to conserve lepton number and spin, an electron is accompanied by an antineutrino $(\bar{\nu})$

(NCERT Ed. 2025-26, 12th, Page No. 310)

22. (4)

Identify Magnetic Field (B): Using the right-hand thumb rule, the magnetic field created by the wire at the position of the rod is directed into the page.

For a conductor moving with velocity v, the direction of the force on positive charges (and thus the higher potential end) is given by the cross product $\vec{v} \times \vec{B}$.

Conclusion: Positive charges accumulate at end A, making its potential higher than that at B.

(NCERT Ed. 2025-26, 12th, Page No. 162)

23. (2)

Condition: $2\pi r = n\lambda \Rightarrow r = n\left(\frac{\lambda}{2\pi}\right)$. Checking options, only Option (2) fits this form where $n = 3$

(NCERT Ed. 2025-26, 12th, Page No. 301)

24. (3)

In a ferromagnetic material, a domain is a region where individual atomic magnetic moments are all aligned in the same direction, resulting in a large non-zero net magnetic moment for each domain.

Statement (3) claims it is zero, which is incorrect.

(NCERT Ed. 2025-26, 12th, Page No. 149)

25. (1)

Time period $T \propto n^3$. Ratio = $(2^3):(3^3) = 8:27$.

(NCERT Ed. 2025-26, 12th, Page No. 299)

26. (1)

According to Lenz's Law, the induced current will flow in a direction that opposes the increase in magnetic flux.

- Analyze the Change: The external magnetic field (B) is into the page ($\times$) and increasing. To oppose this, the induced magnetic field must be out of the page ($\cdot$).

The loop on the right (CD) has a larger area than the loop on the left (AB). Therefore, its induced emf is larger and determines the net direction of current.

- Apply Right-Hand Rule: For the net induced field to be out of the page, the current in the dominant right loop must be counter-clockwise. Tracing this loop, the current flows from D to C.

  Following the continuous wire through the crossing point, a current going from D to C must then flow from B to A in the smaller loop.

(NCERT Ed. 2025-26, 12th, Page No. 160)

27. (4)

Lyman limit ($n = 1$): $\frac{1}{\lambda_L} = R(1)$.

Balmer limit ($n = 2$): $\frac{1}{\lambda_B} = R\left(\frac{1}{2^2}\right) = \frac{R}{4}$.

$\lambda_B = \frac{4}{R} = 4\lambda_L$.

(NCERT Ed. 2025-26, 12th, Page No. 302)

28. (4)

In an electromagnetic wave, the electric field vector ($\vec{E}$) and the magnetic field vector ($\vec{B}$) are mutually perpendicular to each other and to the direction of propagation.

Statement (4) incorrectly claims they are parallel to each other.

(NCERT Ed. 2025-26, 12th, Page No. 206)

29. (3)

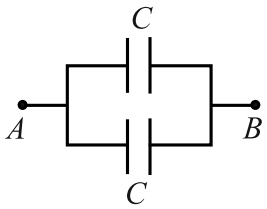
$\lambda \propto \frac{1}{\sqrt{V}}$. Ratio $\frac{\lambda'}{\lambda} = \sqrt{\frac{10}{40}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$.

(NCERT Ed. 2025-26, 12th, Page No. 286)

[Contd...

30. (2)

It is a balanced wheat stone bridge pair.



$C_{eq} = 2C$

(NCERT Ed. 2025-26, 12th, Page No. 71)

31. (4)

$eV = \frac{hc}{\lambda} - \phi$ ...(i)

$e\left(\frac{V}{4}\right) = \frac{hc}{3\lambda} - \phi$ ...(ii)

From (ii),

$eV = \frac{4hc}{3\lambda} - 4\phi.$

Equate with (i):

$\frac{hc}{\lambda} - \phi = \frac{4hc}{3\lambda} - 4\phi \Rightarrow 3\phi = \frac{hc}{3\lambda} \Rightarrow \phi = \frac{hc}{9\lambda}.$

Threshold $\lambda_0 = \frac{hc}{\phi} = 9\lambda.$

(NCERT Ed. 2025-26, 12th, Page No. 281)

32. (1)

- Identify Configuration: The dielectric fills half the space such that the plates are divided by area. This acts as two capacitors connected in parallel.
- Individual Capacitances:

Capacitance of the air half:

$$C_1 = \frac{\epsilon_0\left(\frac{A}{2}\right)}{d} = \frac{C}{2}$$

Capacitance of the dielectric half:

$$C_2 = \frac{K\epsilon_0\left(\frac{A}{2}\right)}{d} = \frac{KC}{2}$$

Equivalent Capacitance (C'):

$$C' = C_1 + C_2 = \frac{C}{2} + \frac{KC}{2} \Rightarrow C' = \frac{C}{2}[1+K]$$

(NCERT Ed. 2025-26, 12th, Page No. 71)

33. (1)

First polaroid transmits

$I_1 = \frac{I_0}{2}$

Second polaroid transmits

$I_2 = I_1\cos^2(45^\circ) = \left(\frac{I_0}{2}\right)\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{I_0}{4}.$

Percentage = 25%.

(NCERT Ed. 2025-26, 12th, Page No. 270)

34. (1)

For the potential difference between B and D to be zero, the Wheatstone bridge must be balanced.

Effective Resistances:

$R_{\text{Top-Left}} = \frac{12\times 4}{12+4} = \frac{48}{16} = 3\Omega$

$R_{\text{Top-Right}} = \frac{x\times 1}{x+1} = \frac{x}{x+1}\Omega$

$R_{\text{Bottom-Left}} = 2+2 = 4\Omega$

$R_{\text{Bottom-Right}} = x\Omega$

Applying Balance Condition: $\frac{3}{\left(\frac{x}{x+1}\right)} = \frac{4}{x}$

$\frac{3(x+1)}{x} = \frac{4}{x}$

$3x = 1 \Rightarrow x = \frac{1}{3}\Omega$

Comparing $x = \frac{1}{3}$ with $x = \frac{1}{n}$, we get $n = 3$.

(NCERT Ed. 2025-26, 12th, Page No. 99)

35. (2)

For minimum deviation, $r = A/2 = 30^\circ$.

Using Snell's Law: $\mu = \frac{\sin i}{\sin r} \Rightarrow \sqrt{3} = \frac{\sin i}{\sin 30^\circ}.$

$\sin i = \sqrt{3}\times 0.5 = \frac{\sqrt{3}}{2} \Rightarrow i = 60^\circ.$

(NCERT Ed. 2025-26, 12th, Page No. 238)

[Contd...

36. (3)

Flux through total sphere = q/ϵ_0. Flux through hemisphere = $\frac{1}{2}\frac{q}{\epsilon_0} = \frac{17.7\times10^{-12}}{2\times8.85\times10^{-12}} = \frac{2}{2} = 1$.

(NCERT Ed. 2025-26, 12th, Page No. 29)

37. (4)

Charge flow $\Delta q = \frac{\Delta\phi}{R} = \frac{50-10}{10} = \frac{40}{10} = 4$ C.

(Time is irrelevant for total charge).

(NCERT Ed. 2025-26, 12th, Page No. 156)

38. (1)

$Q = (\frac{e}{m}) \times \text{mass} = (1.76\times10^{11}) \times (1\times10^{-6}\text{kg})$

$= 1.76\times10^{5}$ C.

(NCERT Ed. 2025-26, 12th, Page No. 2)

39. (2)

Impedance $Z = \frac{V_0}{I_0} = \frac{100}{4} = 25\Omega$.

(NCERT Ed. 2025-26, 12th, Page No. 183)

40. (1)

Total Charge: $Q_{\text{total}} = q + q = 2q$.

When connected, potential $V \propto \frac{Q}{R}$ must be same.

Thus, final charges are shared in the ratio of radii (R: 2R) Final charge on Small Sphere

$q_{\text{small}} = \left(\frac{R}{R+2R}\right)\times 2q = \frac{2q}{3}$.

Charge Transferred: Initial charge was q.

Charge that crossed the switch $\Delta q = q - \frac{2q}{3} = \frac{q}{3}$.

(NCERT Ed. 2025-26, 12th, Page No. 50)

41. (3)

$V_{\text{center}} = 1.5\frac{kQ}{R} = \frac{3kQ}{2R}$.

Required $V = \frac{1}{3}V_{\text{center}} = \frac{kQ}{2R}$.

For outside points, $V = \frac{kQ}{r}$.

Equating: $\frac{kQ}{r} = \frac{kQ}{2R} \Rightarrow r = 2R$.

(NCERT Ed. 2025-26, 12th, Page No. 53)

42. (4)

Net magnetic force on point charge q is

$\vec{F}_B = q(\vec{v}\times\vec{B})$

Since $\vec{F}_B \neq 0$ So, $vB\sin\theta \neq 0$

$\Rightarrow \sin\theta \neq 0$

(NCERT Ed. 2025-26, 12th, Page No. 110)

43. (1)

$B_0 = \frac{E_0}{c} = \frac{60}{3\times10^8} = 20\times10^{-8}$

$= 2\times10^{-7}$ T.

(NCERT Ed. 2025-26, 12th, Page No. 207)

44. (3)

Power $P = \frac{V^2}{R} \Rightarrow R \propto V^2$ (for same power rating).

Ratio: $\frac{R_{\text{USA}}}{R_{\text{India}}} = \left(\frac{V_{\text{USA}}}{V_{\text{India}}}\right)^2 = \left(\frac{110}{220}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

$R_{\text{USA}} = \frac{R}{4}$.

(NCERT Ed. 2025-26, 12th, Page No. 108)

45. (2)

When wire A is stretched 2 $\times$, its new resistance $R'_A = n^2R_A = 4R_A$.

Given $R'_A = R_B \Rightarrow 4R_A = R_B$.

Since $R \propto L$ (for same area initially),

$4L_A = L_B$.

Ratio $L_A : L_B = 1:4$.

(NCERT Ed. 2025-26, 12th, Page No. 94)

[Contd...

46. (1)

Cannizzaro reaction: Aldehyde which do not have an α-H atom, undergo disproportionation on heating with alkali to form an alcohol and carboxylic acid salt.

$$2PhCHO \xrightarrow[\Delta]{\text{Conc. KOH}} PhCH_2OH + PhCOOK$$

(NEW NCERT 12th Part-II Page No. 242)

47. (4)

- $[Mn_2(CO)_{10}]$ is a organometallic compound as it contain metal-carbon bond.
- Metallic carbonyl involve synergic bonding.

(NEW NCERT 12th Part-I Page No. 136)

48. (4)

Hydrolysis of ethyl acetate in acidic medium is a pseudo first order reaction.

(NEW NCERT 12th Part-I Page No. 78)

49. (3)

$$CH_3 - COO^- + Fe^{3+} + H_2O \rightarrow \underset{\text{brown red}}{[Fe(OH)_2(CH_3COO)]}$$

50. (1)

- Higher the E°_{OP}, more will be the reducing power.
- E°_{OP}: Al > Fe > Cu > Ag

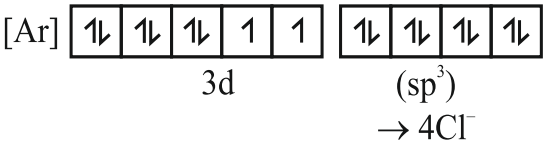
(NEW NCERT 12th Part-I Page No. 37)

51. (4)

Acidic strength: $H_2O < H_2S < H_2Se < H_2Te$

52. (4)

Ni^{2+} in $[Ni(H_2O)_4]^{2+}$:



No. of unpaired $e^- = 2$

$$\mu = \sqrt{2(2+2)}\ BM = \sqrt{8}\ BM$$

(NEW NCERT 12th Part-I Page No. 130)

53. (1)

Sc shows only +3 oxidation state.

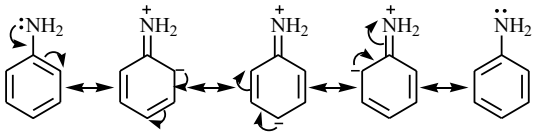
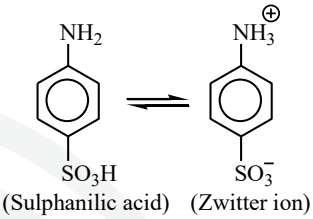
(NEW NCERT 12th Part-I Page No. 96)

54. (2)

- Higher the value of ($i \times M$), higher will be the osmotic pressure of aqueous solution.
- Solute: NaCl $K_4[Fe(CN)_6$ $CaCl_2$ Glucose
  i : 2 5 3 1

(NEW NCERT 12th Part-I Page No. 24)

55. (3)

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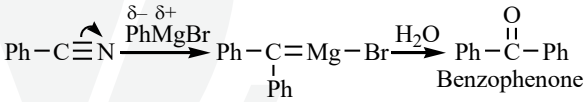
(NEW NCERT 12th Part-II Page No. 269)

56. (3)

Histidine is an essential amino acid.

(NEW NCERT 12th Part-II Page No. 291)

57. (3)



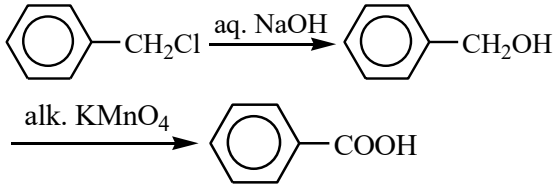
(NEW NCERT 12th Part-II Page No. 234)

58. (1)

Given compound is α-D-(–)-Fructofuranose

(NEW NCERT 12th Part-II Page No. 286)

59. (1)



(NEW NCERT 12th Part-II Page No. 210)

60. (4)

S_N2 reactions are favoured in polar aprotic solvent.

(NEW NCERT 12th Part-II Page No. 171)

61. (2)

$(-\Delta_{eq}H^\circ)$: Cl > F > Br > I

[Contd...

62. (2)

$FeSO_4.(NH_4)_2SO_4.6H_2O$ is a double salt.

(NEW NCERT 12th Part-I Page No. 120)

63. (2)

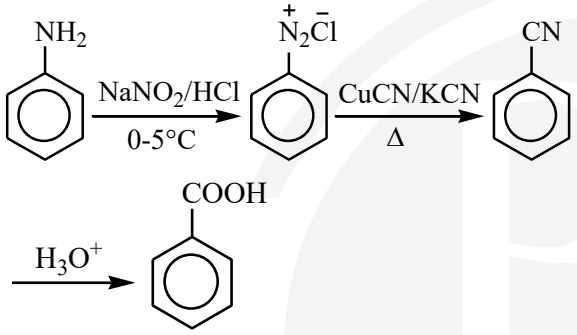
$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A_t]}$$

$$\frac{0.693}{t_{1/2}} = \frac{2.303}{1}\log\left(\frac{100}{10}\right)$$

$$t_{1/2} = \frac{0.693}{2.303} = 0.3\ \text{min}$$

(NEW NCERT 12th Part-I Page No. 73)

64. (2)



(NEW NCERT 12th Part-II Page No. 275)

65. (1)

$$\overset{+6}{Mn}O_4^{2-} \longrightarrow \overset{+7}{Mn}O_4^- + 1e^-$$

1 mol 1F

Charge required = 0.1 × 96500 C = 9650 C

(NEW NCERT 12th Part-I Page No. 52)

66. (4)

Due to almost identical radii (properties), Zr and Hf are called twins elements.

(NEW NCERT 12th Part-I Page No. 109)

67. (2)

The approximate composition of a azeotrope of C_2H_5OH and H_2O by volume is 95% C_2H_5OH and 5% H_2O.

(NEW NCERT 12th Part-I Page No. 15)

68. (4)

- B_{12} vitamin can be stored in our body.
- Lactose is a reducing sugar.

(NEW NCERT 12th Part-II Page No. 296)

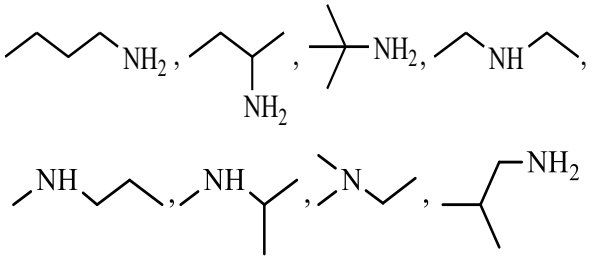
69. (3)

Reactivity of alcohols towards Lucas reagent:

$3° > 2° > 1°$

(NEW NCERT 12th Part-II Page No. 208)

70. (4)



(NEW NCERT 12th Part-II Page No. 262)

71. (1)

First inert compound is formed by Neil Bartlett by reacting Xenon with PtF_2.

72. (2)

In case of symmetrical alkene peroxide effect is not observed.

(NEW NCERT 12th Part-II Page No. 165)

73. (2)

$$r = k[A]^{n_1}[B]^{n_2}$$

$$1.5\times10^{-4} = k[0.1]^{n_1}[0.2]^{n_2} \quad \ldots (1)$$

$$1.5\times10^{-4} = k[0.2]^{n_1}[0.2]^{n_2} \quad \ldots (2)$$

$$3\times10^{-4} = k[0.2]^{n_1}[0.4]^{n_2} \quad \ldots (3)$$

From (1) and (2), $\frac{1}{1} = \left(\frac{1}{2}\right)^{n_1} \Rightarrow n_1 = 0$

From (2) and (3), $\frac{1}{2} = \left(\frac{1}{2}\right)^{n_2} \Rightarrow n_2 = 1$

$\therefore$ overall order = 0 + 1 = 1

(NEW NCERT 12th Part-I Page No. 68)

74. (3)

$$K_2[HgI_4] + KOH \xrightarrow{NH_3} \underset{\text{brown ppt}}{Hg_2O\cdot NH_2, I}$$

75. (2)

For Daniell cell, $E^\circ_{cell} = 1.1\,V$

(NEW NCERT 12th Part-I Page No. 32)

[Contd...

76. (4)

Actinoid magnetic properties are more complex than lanthanoids.

(NEW NCERT 12th Part-I Page No. 112, 113)

77. (4)

More the value of K_H, lesser will be the solubility of gas in water.

(NEW NCERT 12th Part-I Page No. 07)

78. (3)

Ethanol is dehydrated to ethene in the presence of H_2SO_4 at 443 K. At 413 K, ethoxy ethane is the main product.

(NEW NCERT 12th Part-II Page No. 215)

79. (4)

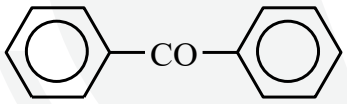


(NEW NCERT 12th Part-II Page No. 240)

80. (2)

Reactivity order for nucleophilic addition:

$HCHO > CH_3CHO > CH_3COCH_3 >$



(NEW NCERT 12th Part-II Page No. 236)

81. (4)

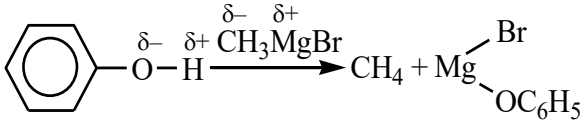
- Nitrogen is restricted to a maximum covalency of four.
- Hydride NH_3 PH_3 AsH_3 SbH_3 BiH_3
  BP(K) 238.5 185.5 210.6 254.6 290

82. (2)

Fluoroalkane can be prepared by Swarts reaction.

(NEW NCERT 12th Part-II Page No. 166)

83. (2)



(NEW NCERT 12th Part-II Page No. 181)

84. (1)

$$p_{Benzene} = 80 \times \frac{2}{3} \text{ torr}$$

$$p_{Toluene} = 24 \times \frac{1}{3} \text{ torr}$$

$$Y_{Toluene} = \frac{p_{Toluene}}{p_{Total}} = \frac{24 \times \frac{1}{3}}{80 \times \frac{2}{3} + 24 \times \frac{1}{3}} = 0.13$$

(NEW NCERT 12th Part-I Page No. 11)

85. (1)

Complexes in which a metal is bound to only one kind of donor groups known as homoleptic.

(NEW NCERT 12th Part-I Page No. 122)

86. (2)

For n^{th} order reaction, unit of k is $(\text{mol } L^{-1})^{1-n} s^{-1}$.

(NEW NCERT 12th Part-I Page No. 69)

87. (2)

For weak electrolytes, specific conductance decreases on dilution.

(NEW NCERT 12th Part-I Page No. 49)

88. (2)

- Zn^{2+}: Colourless
- Mn^{2+}: Pink
- Cu^{2+}: Blue
- Fe^{2+}: Green

(NEW NCERT 12th Part-I Page No. 103)

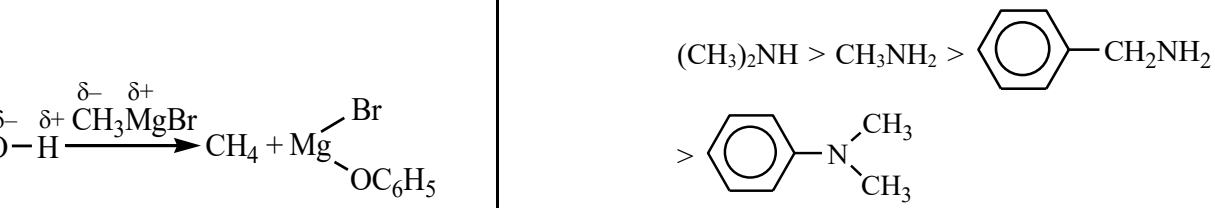
89. (4)

During denaturation of protein primary structure remain intact.

(NEW NCERT 12th Part-II Page No. 293)

90. (1)

- Lesser the pK_b value, more will be the basic nature of amine.
- Basic nature:



(NEW NCERT 12th Part-II Page No. 267)

[Contd...

91. **(1)**
Panchanan Maheshwari emphasised the need for initiation of work on artificial culture of immature embryos. His work on test tube fertilisation and intra-ovarian pollination won worldwide acclaim.
(NCERT Ed. 2025-26, 12th, Page No. 2)

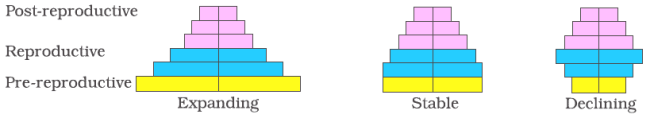
92. **(3)**
More than 70 per cent of all the species recorded are animals, while plants (including algae, fungi, bryophytes, gymnosperms and angiosperms) comprise no more than 22 per cent of the total.
(NCERT Ed. 2025-26, 12th, Page No. 218)

93. **(1)**
Decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and water-soluble substances like sugars.
(NCERT Ed. 2025-26, 12th, Page No. 208)

94. **(4)**
The sex determination in honey bee is based on the number of sets of chromosomes an individual receives. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilised egg develops as a male (drone) by means of parthenogenesis. This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes. This is called as haplodiploid sex-determination system and has special characteristic features such as the males produce sperms by mitosis, they do not have father and thus cannot have sons, but have a grandfather and can have grandsons.
(NCERT Ed. 2025-26, 12th, Page No. 71)

95. **(2)**
DNA is a long polymer of deoxyribonucleotides. 2'-OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable. RNA is also now known to be catalytic, hence reactive. Therefore, DNA chemically is less reactive and structurally more stable when compared to RNA. Therefore, among the two nucleic acids, the DNA is a better genetic material. In fact, the presence of thymine at the place of uracil also confers additional stability to DNA.
(NCERT Ed. 2025-26, 12th, Page No. 87)

96. **(3)**
A population age pyramid with a broad base that narrows towards the apex reflects growing population.



(NCERT Ed. 2025-26, 12th, Page No. 191)

97. **(1)**
If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct for lack of food. This is the reason why predators in nature are 'prudent'.
(NCERT Ed. 2025-26, 12th, Page No. 198)

98. **(3)**
The repressor of the *lac* operon is synthesised (all-the-time – constitutively) from the *i* gene. The *lac* operon consists of one regulatory gene (the *i* gene – here the term *i* does not refer to inducer, rather it is derived from the word inhibitor) and three structural genes (*z*, *y*, and *a*).
(NCERT Ed. 2025-26, 12th, Page No. 101)

99. **(2)**
The disease shown in the given pedigree cannot be Y-linked since females are also affected. It cannot be a dominant trait since carrier females are also produced. It cannot be autosomal recessive because in that case all the F_3 progeny would be carriers. Therefore, the trait is X-linked recessive.
(NCERT Ed. 2025-26, 12th, Page No. 73)

100. **(3)**
Chromosome 1 has most genes (2968), and the Y has the fewest (231).
(NCERT Ed. 2025-26, 12th, Page No. 104)

101. **(2)**
Species composition and stratification are the two main structural features of an ecosystem. Productivity, decomposition, energy flow, and nutrient cycling are the four important components of an ecosystem.
(NCERT Ed. 2025-26, 12th, Page No. 206)

102. **(2)**
In India, ecologically unique and biodiversity-rich regions are legally protected as biosphere reserves, national parks and sanctuaries. India now has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries. Three of the hotspots – Western Ghats and Sri Lanka, Indo-Burma and Himalaya – cover our country's exceptionally high biodiversity regions.
(NCERT Ed. 2025-26, 12th, Page No. 225)

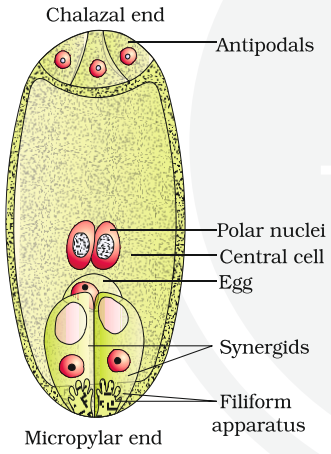
103. (1)

Methanogens are present in the rumen (a part of stomach) of cattle and help in the breakdown of cellulose. The technology of biogas production was developed in India mainly due to the efforts of Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC).

(NCERT Ed. 2025-26, 12th, Page No. 156)

104. (3)

'A' represents the antipodal cells, which are located at the chalazal end of the embryo sac. 'F' represents the filiform apparatus found in the synergids. It is located at the micropylar end. B (Polar nuclei) are haploid (n). D (Egg cell) is a female gamete and is haploid (n). E (Synergid) is a cell of the gametophyte and is haploid (n). 'C' is the Central cell. After double fertilization (fusion with a male gamete), the central cell form the primary endosperm cell, which develops into the endosperm. It is the Egg cell (D) that forms the zygote.



A diagrammatic representation of the mature embryo sac

(NCERT Ed. 2025-26, 12th, Page No. 10)

105. (3)

Visiting flamingoes and resident fishes in some south American lakes: This is competition (-/-). Both compete for the same zooplankton in the lake. Neither benefits from the other's presence; both are negatively impacted.

(NCERT Ed. 2025-26, 12th, Page No. 201)

106. (1)

Turner's Syndrome : Such a disorder is caused due to the absence of one of the X chromosomes, i.e., 45 with X0, Such females are sterile as ovaries are rudimentary besides other features including lack of other secondary sexual characters. Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy. For example, Turner's syndrome results due to loss of an X chromosome in human females. The pattern of inheritance of Mendelian disorders can be traced in a family by the pedigree analysis, but not in chromosomal disorders like Turner's syndrome.

(NCERT Ed. 2025-26, 12th, Page No. 76)

107. (2)

The length of DNA is usually defined as number of nucleotides (or a pair of nucleotide referred to as base pairs) present in it. This also is the characteristic of an organism. For example, a bacteriophage known as ϕ ×174 has 5386 nucleotides. In some viruses tht have RNA as the genetic material, the flow of genetic information is in reverse direction, that is, from RNA to DNA.

(NCERT Ed. 2025-26, 12th, Page No. 80)

108. (1)

Cyclosporin A is produced by the fungus *Trichoderma polysporum*. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. Streptokinase is produced by the bacterium *Streptococcus*. *Saccharomyces cerevisiae* used for bread-making and commonly called brewer's yeast, is used for fermenting malted cereals and fruit juices, to produce ethanol.

(NCERT Ed. 2025-26, 12th, Page No. 153)

109. (3)

In a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits.

(NCERT Ed. 2025-26, 12th, Page No. 20)

110. (4)

Some examples of recent extinctions include the dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia). Passenger pigeon became extinct due to over-exploitation. The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.

(NCERT Ed. 2025-26, 12th, Page No. 221)

111. (1)

Of the incident solar radiation less than 50 per cent of it is photosynthetically active radiation (PAR).

(NCERT Ed. 2025-26, 12th, Page No. 209)

112. (1)

Mendel selected 14 true-breeding pea plant varieties. Tall height is a dominant trait in garden pea.

(NCERT Ed. 2025-26, 12th, Page No. 54)

[Contd...

113. (4)

AUG has dual functions. It codes for Methionine (met), and it also act as initiator codon. UAA, UAG, UGA are stop terminator codons.

(NCERT Ed. 2025-26, 12th, Page No. 96)

114. (3)

Apomixis is a form of asexual reproduction that mimics sexual reproduction.

(NCERT Ed. 2025-26, 12th, Page No. 22)

115. (3)

Many members of the genus *Glomus* form mycorrhiza.

(NCERT Ed. 2025-26, 12th, Page No. 158)

116. (1)

The steps of DNA fingerprinting are;
(i) isolation of DNA, (ii) digestion of DNA by restriction endonucleases, (iii) separation of DNA fragments by electrophoresis, (iv) transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon, (v) hybridisation using labelled VNTR probe, and (vi) detection of hybridised DNA fragments by autoradiography.

(NCERT Ed. 2025-26, 12th, Page No. 106)

117. (2)

Secondary treatment or Biological treatment : The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs.

(NCERT Ed. 2025-26, 12th, Page No. 154)

118. (3)

The O blood group is recessive and only occurs with the genotype *ii*. For a child to have blood group O, they must inherit one *i* allele from each parent. Since none of the parents have O blood group as per the given question, it is possible only in option 3.

(NCERT Ed. 2025-26, 12th, Page No. 61)

119. (2)

Baculoviruses are pathogens that attack insects and other arthropods. The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*. These viruses are excellent candidates for species-specific, narrow spectrum insecticidal applications. They have been shown to have no negative impacts on plants, mammals, birds, fish or even on non-target insects. This is especially desirable when beneficial insects are being conserved to aid in an overall integrated pest management (IPM) programme, or when an ecologically sensitive area is being treated.

(NCERT Ed. 2025-26, 12th, Page No. 157)

120. (1)

Species A	Species B	Name of Interaction
+	+	*Mutualism*
–	–	*Competition*
+	–	*Predation*
+	–	*Parasitism*
+	0	*Commensalism*
–	0	*Amensalism*

Population interactions

(NCERT Ed. 2025-26, 12th, Page No. 197)

121. (4)

In snapdragon (*Antirrhinum* sp.), although the phenotypic ratio in F_2 deviates from Mendelian monohybrid cross (it becomes 1 : 2 : 1 instead of 3 : 1), the genotypic ratio remains 1 : 2 : 1, which is the same as Mendelian inheritance. Hence, the genotypic ratio does not deviate.

(NCERT Ed. 2025-26, 12th, Page No. 60)

122. (1)

The zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo.

(NCERT Ed. 2025-26, 12th, Page No. 19)

123. (1)

Nucellus cells are diploid like MMC, megaspore is haploid.

(NCERT Ed. 2025-26, 12th, Page No. 10)

124. (1)

Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.

(NCERT Ed. 2025-26, 12th, Page No. 199)

125. (2)

The generative cell of a pollen grain is small and floats in the cytoplasm of the vegetative cell.

(NCERT Ed. 2025-26, 12th, Page No. 7)

[Contd...

126. (2)

Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.

(NCERT Ed. 2025-26, 12th, Page No. 67)

127. (2)

The amount of available energy is maximum at producer's level i.e. T_1.

(NCERT Ed. 2025-26, 12th, Page No. 211)

128. (1)

Saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem. Pyramid of energy is always upright, can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.

(NCERT Ed. 2025-26, 12th, Page No. 214)

129. (4)

In bacteria, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the translation can begin much before the mRNA is fully transcribed. Consequently, the transcription and translation can be coupled in bacteria. The transcription is catalysed by DNA-dependent RNA polymerase and translation by ribozyme.

(NCERT Ed. 2025-26, 12th, Page No. 94)

130. (4)

Phytoplanktons are the producers in a pond ecosystem.

(NCERT Ed. 2025-26, 12th, Page No. 206)

131. (3)

In general, species diversity decreases as we move away from the equator towards the poles. With very few exceptions, tropics (latitudinal range of 23.5° N to 23.5° S) harbour more species than temperate or polar areas. Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds. A forest in a tropical region like Equador has up to 10 times as many species of vascular plants as a forest of equal area in a temperate region like the Midwest of the USA. The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth.

(NCERT Ed. 2025-26, 12th, Page No. 220)

132. (2)

The family pedigree of Queen Victoria shows a number of haemophilic descendents as she was a carrier of the disease. In addition to the above, mutation also arise due to change in a single base pair of DNA. This is known as point mutation. A classical example of such a mutation is sickle cell anemia. Phenylketonuria is a pleiotropic autosomal recessive disorder. Colour blindness is a sex-linked recessive disorder due to defect in either red or green cone of eye resulting in failure to discriminate between red and green colour. This defect is due to mutation in certain genes present in the X chromosome.

(NCERT Ed. 2025-26, 12th, Page No. 74)

133. (1)

Habitat loss and fragmentation: This is the most important cause driving animals and plants to extinction.

(NCERT Ed. 2025-26, 12th, Page No. 222)

134. (2)

Francis Crick won Nobel prize in 1962.

(NCERT Ed. 2025-26, 12th, Page No. Unit VII Intro)

135. (4)

Cleistogamous flowers (found in plants like *Viola*, *Oxalis*, and *Commelina*) do not open at all. Because the flower remains closed, cross-pollination or pollination from other flowers on the same plant is physically impossible. Therefore, the only possible method is self-pollination within that closed bud (autogamy). In the open chasmogamous flowers, the reproductive organs are exposed. While they can self-pollinate, geitonogamy as well as xenogamy is possible in them. Papaya plants are typically dioecious, meaning male and female flowers grow on separate plants. Since a single plant cannot fertilize itself (no male and female parts on the same plant), pollen must come from a different plant. This transfer of pollen between genetically different plants is called Xenogamy.

(NCERT Ed. 2025-26, 12th, Page No. 12)

[Contd...

136. (4)

Seminal plasma is the fluid part of semen formed mainly by secretions of seminal vesicles, prostate gland and bulbourethral glands. These secretions provide nutrition (like fructose), alkalinity and a medium for sperm movement.

(NCERT Ed. 2025-26, 12th, Page No. 28)

137. (3)

A biopsy means removing a small piece of suspected tissue and examining it under a microscope (histopathology). This method directly confirms cancer by showing abnormal cells in the tissue sample.

(NCERT Ed. 2025-26, 12th, Page No. 141)

138. (4)

List-I	List-II
Ti-plasmid	Vector derived from *Agrobacterium tumefaciens*
Retrovirus	Used to deliver genes into animal cells
pBR322	Specifically engineered cloning vector for *E. coli*
Bacteriophage	Virus that infects bacteria and used as vectors

(NCERT Ed. 2025-26, 12th, Page No. 170)

139. (1)

Divergent evolution is shown by homologous organs (same origin but different functions). Forelimbs of cheetah and human, thorns of *Bougainvillea* and tendrils of *Cucurbita* (both are stem modifications), and vertebrate hearts/brains share common origin, so all these represent divergent evolution.

(NCERT Ed. 2025-26, 12th, Page No. 114)

140. (2)

Anamnestic response means secondary immune response. When the same pathogen enters again, memory cells respond rapidly and strongly, producing a highly intensified response compared to the primary response.

(NCERT Ed. 2025-26, 12th, Page No. 135)

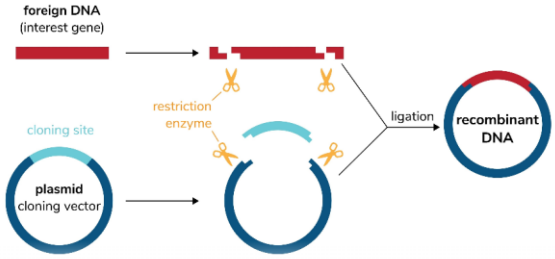
141. (3)

Many recombinant therapeutics are made to be exactly like human proteins, so the body usually does not treat them as foreign. Reason is false because these products are not purified from animal sources and hence reduce unwanted immune reactions.

(NCERT Ed. 2025-26, 12th, Page No. 181)

142. (4)

In recombinant DNA technology, first DNA is cut at specific sites by a restriction endonuclease (process A). Then the desired fragment is joined to vector DNA by DNA ligase (process B), forming recombinant DNA.



(NCERT Ed. 2025-26, 12th, Page No. 167)

143. (2)

During the follicular (proliferative) phase, estrogen released by growing follicles repairs and regenerates the endometrium after menstruation. Progesterone mainly acts in the luteal (secretory) phase.

(NCERT Ed. 2025-26, 12th, Page No. 34)

144. (3)

Menarche is the first menstruation at puberty. The luteal phase is also called the secretory phase.

(NCERT Ed. 2025-26, 12th, Page No. 33)

145. (1)

Transgenic mice are widely used as experimental models to test the safety of vaccines, including polio vaccine, before use in humans. They help in checking immune response and possible side effects.

(NCERT Ed. 2025-26, 12th, Page No. 184)

146. (2)

In rural areas, pollution was low, so lichens continued to grow on tree trunks, keeping the bark light in colour. Therefore, light-coloured moths stayed camouflaged and melanic moths did not get a selective advantage, so melanic count remained low.

(NCERT Ed. 2025-26, 12th, Page No. 116)

147. (4)

List-I	List-II
Golden rice	Vitamin A enriched
Bt Cotton	Pest resistance
Pomato	Somatic hybrid
Somaclones	Genetically identical plants produced via tissue culture

(NCERT Ed. 2025-26, 12th, Page No. 179)

[Contd...

148. (4)

Flying squirrel (placental mammal) and flying phalanger (Australian marsupial) show convergent evolution because both developed similar gliding adaptations independently for the same lifestyle.

(NCERT Ed. 2025-26, 12th, Page No. 118)

149. (1)

The 'family planning' programmes were periodically assessed over the past decades. Improved programmes covering wider reproductive related areas are currently in operation under the popular name RCH programmes.

(NCERT Ed. 2025-26, 12th, Page No. 42)

150. (2)

Immediately after implantation, the inner cell mass (embryo) differentiates into an outer layer called ectoderm and an inner layer called endoderm. A mesoderm soon appears between the ectoderm and the endoderm. These three layers give rise to all tissues (organs) in adults.

(NCERT Ed. 2025-26, 12th, Page No. 38)

151. (3)

Biotechnology needs regulation for safety, ethics and environmental concerns. Statement II is also correct because Humulin produced using recombinant methods has the same structure as human insulin and it performs the same biological function as human insulin.

(NCERT Ed. 2025-26, 12th, Page No. 184)

152. (1)

Ascariasis is caused by the intestinal endoparasite Ascaris lumbricoides (a roundworm). It is a water-borne/food-borne disease that spreads through the ingestion of water, vegetables, or fruits contaminated with the eggs of the parasite, which are excreted in the feces of infected individuals. There is no requirement for an insect vector in its life cycle.

(NCERT Ed. 2025-26, 12th, Page No. 133)

153. (2)

List-I	List-II
Acrosome	Enzymes for fertilisation
Mitochondria	Energy source for sperm movement
Fimbriae	Collection of ovum
Alveoli	Secretion of milk

(NCERT Ed. 2025-26, 12th, Page No. 32)

154. (2)

A stirred-tank bioreactor is usually cylindrical with a curved base to allow proper mixing and aeration.

(NCERT Ed. 2025-26, 12th, Page No. 174)

155. (3)

The method of directly injecting a sperm into an ovum *in vitro* is called ICSI (Intracytoplasmic Sperm Injection).

(NCERT Ed. 2025-26, 12th, Page No. 48)

156. (4)

- Complications from various Sexually Transmitted Infections (STIs) may lead to pelvic inflammatory diseases (PID), stillbirths, and ectopic pregnancies.
- These serious health issues underscore the importance of preventing and treating STIs to safeguard reproductive health and overall well-being.

(NCERT Ed. 2025-26, 12th, Page No. 47)

157. (1)

List-I	List-II
Opioids	*Papaver somniferum*
Cannabinoids	*Cannabis sativa*
Cocaine	*Erythroxylum coca*
Hallucinogens	*Datura*

(NCERT Ed. 2025-26, 12th, Page No. 143)

158. (2)

EFB definition includes both traditional uses (like fermentation) and modern molecular biotechnology (like genetic engineering). So it gives a broader meaning, covering old and new biotechnology.

(NCERT Ed. 2025-26, 12th, Page No. 163)

159. (2)

Eli Lilly successfully produced human insulin in 1983 by producing A and B chains separately in *E. coli* and then combining them.

(NCERT Ed. 2025-26, 12th, Page No. 182)

160. (1)

Assertion is true because DNA fragments separate in gel mainly by size. Reason is also true and explains it correctly because agarose gel acts like a molecular sieve, so smaller DNA fragments move faster and travel farther.

(NCERT Ed. 2025-26, 12th, Page No. 168)

[Contd...

161. (3)

The Intra Uterine Devices are presently available as the non-medicated IUDs (e.g., Lippes loop), copper releasing IUDs (CuT, Cu7, Multiload 375) and the hormone releasing IUDs (Progestasert, LNG-20).

(NCERT Ed. 2025-26, 12th, Page No. 44)

162. (3)

MALT forms about 50% of the total lymphoid tissue in the human body. It is present in mucosal linings like gut, respiratory tract, etc., providing first-line immune protection.

(NCERT Ed. 2025-26, 12th, Page No. 138)

163. (4)

RNA interference occurs in many eukaryotes as a defense mechanism and it causes silencing of specific mRNA by complementary RNA.

(NCERT Ed. 2025-26, 12th, Page No. 180)

164. (1)

Vasectomy involves the cutting and tying of vas deferens in males so that gamete transport can be blocked. Oviducts are cut and tied in tubectomy which is a sterilisation procedure for females.

(NCERT Ed. 2025-26, 12th, Page No. 46)

165. (4)

List-I (Embryo development stage)	List-II (Timeline)
Heart formation	After 1 month
Limbs and digits	End of 2nd month
Major organ systems	End of 12 weeks
Eyelashes formed	End of 24 weeks

(NCERT Ed. 2025-26, 12th, Page No. 38)

166. (2)

Smoking increases carbon monoxide (CO) in blood. CO binds haemoglobin very strongly to form carboxyhaemoglobin, reducing oxygen carrying capacity, so haem-bound oxygen decreases.

(NCERT Ed. 2025-26, 12th, Page No. 144)

167. (3)

In PCR, the enzyme used is *Taq* polymerase, which is isolated from the thermophilic bacterium Thermus aquaticus. This enzyme is specifically chosen because it remains stable and functional at the high temperatures (around 94-96°C) required to denature the DNA strands. PCR amplifies DNA, not proteins, and the polymerase is heat-stable, not heat-labile. The correct sequence of steps is Denaturation, Annealing and Extension. Primers are short, single-stranded oligonucleotides, not double-stranded.

(NCERT Ed. 2025-26, 12th, Page No. 173)

168. (4)

Prenatal test called amniocentesis is generally used to find genetic disorders in the growing foetus.

This test involves acquiring a tiny portion of the amniotic fluid that surrounds the foetus in the womb.

In addition to dissolved materials, this fluid contains foetal cells.

The analysis of the foetal cells reveals whether the foetus has any genetic disorders, such as phenylketonuria, based on the chromosomal pattern of the foetal cells.

The procedure is also illegally often used ascertain the gender of the foetus, that has been associated to an increase in female foeticides.

(NCERT Ed. 2025-26, 12th, Page No. 42)

169. (1)

List-I	List-II
First cellular forms of life	2000 mya
Invertebrates formed	500 mya
Jawless fish evolved	350 mya
Sea weeds and few plants	320 mya

(NCERT Ed. 2025-26, 12th, Page No. 122)

170. (3)

The human male ejaculates about 200 to 300 million sperms during a coitus of which, for normal fertility, at least 60 per cent sperms must have normal shape and size and at least 40 per cent of them must show vigorous motility.

(NCERT Ed. 2025-26, 12th, Page No. 32)

171. (2)

The embryos of all vertebrates including human develop a row of vestigial gill slit just behind the head. This shows embryological evidence.

(NCERT Ed. 2025-26, 12th, Page No. 113)

172. (1)

Sporozoites are injected from mosquito salivary glands into humans, and rupture of RBCs releases haemozoin which causes fever and chills; sexual reproduction and fertilization occur in mosquito, not in humans.

(NCERT Ed. 2025-26, 12th, Page No. 131)

[Contd...

173. (1)

Miller's experiment simulated primitive earth atmosphere using CH_4, H_2, NH_3 and water vapour with high temperature and electric discharge (lightning). This setup produced organic molecules like amino acids.

(NCERT Ed. 2025-26, 12th, Page No. 111)

174. (2)

List-I	List-II
GEAC	Validity of GM research
Biopiracy	Unauthorized use of bio-resources
Basmati Patent by American company	US Patent and Trademark Office
Patent terms of emergency provisions	2nd amendment of Indian Patents Bill

(NCERT Ed. 2025-26, 12th, Page No. 185)

175. (3)

Alpha-1-antitrypsin is used for treating emphysema because it protects lung tissue from damage by inhibiting protease enzymes. Lack of this protein can lead to lung tissue destruction and emphysema.

(NCERT Ed. 2025-26, 12th, Page No. 184)

176. (1)

In a tertiary follicle, the fluid-filled cavity is called the antrum. It enlarges as the follicle matures toward Graafian follicle stage.

(NCERT Ed. 2025-26, 12th, Page No. 32)

177. (3)

If recessive allele frequency q = 0.4, then p = 1 – q = 0.6. Heterozygous frequency is $2pq = 2 \times 0.6 \times 0.4 = 0.48$

(NCERT Ed. 2025-26, 12th, Page No. 121)

178. (3)

The first recombinant DNA molecule was constructed using a native plasmid of *Salmonella typhimurium* in early recombinant DNA experiments.

(NCERT Ed. 2025-26, 12th, Page No. 164)

179. (3)

In severe pneumonia, lungs cannot oxygenate blood properly, causing cyanosis. So, lips and fingernails may turn grayish to bluish, which indicates low oxygen levels.

(NCERT Ed. 2025-26, 12th, Page No. 131)

180. (2)

Origin of replication is where DNA replication starts, not translation. Statement II is correct because antibiotic resistance genes (like kanamycin resistance) help in selecting transformed cells.

(NCERT Ed. 2025-26, 12th, Page No. 169)

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